Class 14: Pathway Analysis from RNA-Seq Results

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Overview

In this lab, we analyze RNA-seq data from the GSE37704 dataset. This experiment studies lung fibroblasts after HOXA1 knockdown. We will use DESeq2 to find differentially expressed

genes, then use KEGG, Gene Ontology, and Reactome pathway analysis to understand the biological meaning of the gene expression changes.

Section 1. Differential Expression Analysis

Load DESeq2.

```
library(DESeq2)
```

Load our data files.

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

colData <- read.csv(metaFile, row.names = 1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
countData <- read.csv(countFile, row.names = 1)
head(countData)
```

```
                length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
                SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

The count data has an extra `length` column that does not match the metadata samples, so we need to remove it.

Q1. Complete the code to remove the troublesome first column from countData

```
countData <- as.matrix(countData[, -1])
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
ENSG00000278566         0         0         0         0         0         0
ENSG00000273547         0         0         0         0         0         0
ENSG00000187634       124       123       205       207       212       258
```

I removed the first column using `-1` because the first column is the gene length column, not a sample count column.

Q2. Filter countData to exclude genes where we have 0 read count across all samples

```
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

`rowSums(countData)` adds the counts across all samples for each gene. If the row sum is 0, that gene has no reads in any sample, so it should be removed.

Running DESeq2

Now we can set up the DESeqDataSet object and run the DESeq2 pipeline.

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~ condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```
dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
  ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

Get the DESeq2 results.

```r
res <- results(dds)
res
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 15975 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248215   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630156 1.43993e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
...                   ...            ...       ...        ...         ...
ENSG00000273748  35.30265       0.674387  0.303666   2.220817 2.63633e-02
ENSG00000278817   2.42302      -0.388988  1.130394  -0.344118 7.30758e-01
ENSG00000278384   1.10180       0.332991  1.660261   0.200565 8.41039e-01
ENSG00000276345  73.64496      -0.356181  0.207716  -1.714752 8.63908e-02
ENSG00000271254 181.59590      -0.609667  0.141320  -4.314071 1.60276e-05
                      padj
                 <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76553e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
...                     ...
ENSG00000273748 4.79091e-02
ENSG00000278817 8.09772e-01
ENSG00000278384 8.92654e-01
ENSG00000276345 1.39761e-01
ENSG00000271254 4.53647e-05
```

Check the result names.

```r
resultsNames(dds)
```

```
[1] "Intercept"                           "condition_hoxa1_kd_vs_control_sirna"
```

Q3. Call summary() on the DESeq2 results

```
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

The summary shows how many genes are up-regulated and down-regulated at the default adjusted p-value cutoff of 0.1. This helps us quickly see that HOXA1 knockdown caused many gene expression changes.

Volcano Plot

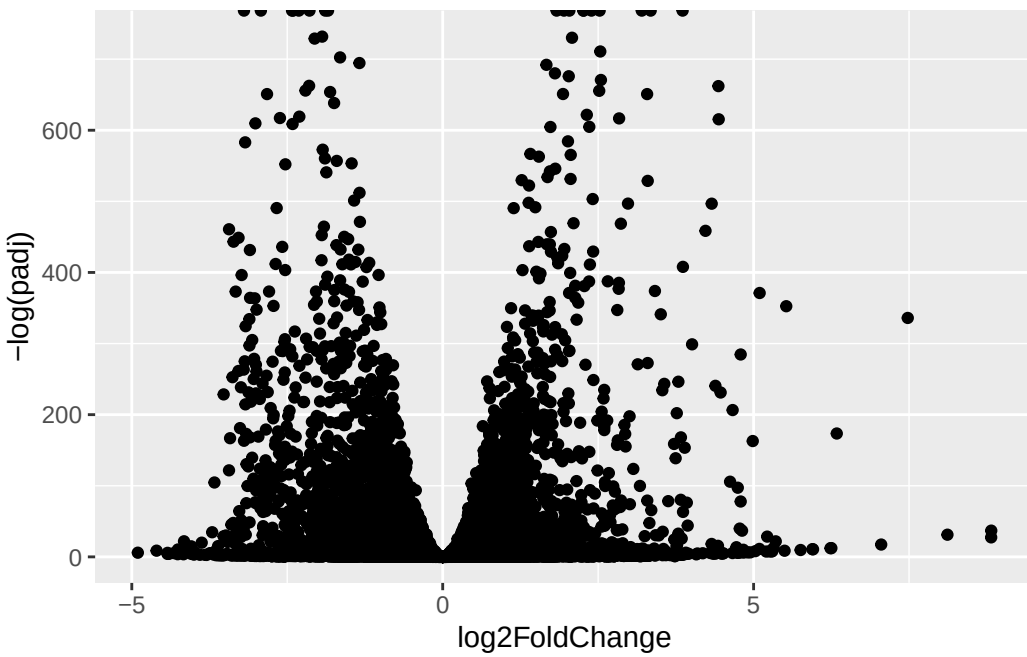
A volcano plot shows log2 fold change on the x-axis and statistical significance on the y-axis.

```
library(ggplot2)
```

Make a basic volcano plot.

```
res_df <- as.data.frame(res)

ggplot(res_df) +
  aes(x = log2FoldChange,
      y = -log(padj)) +
  geom_point()
```

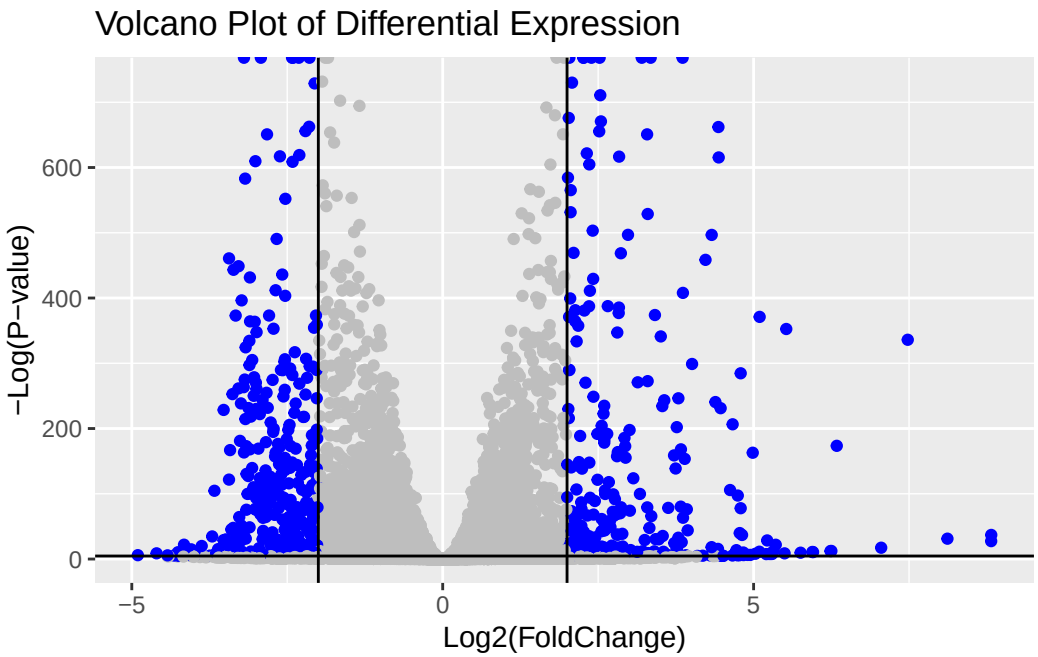
```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Q4. Improve the volcano plot with color, axis labels, and cutoff lines

```
mycols <- rep("gray", nrow(res_df))

mycols[abs(res_df$log2FoldChange) > 2] <- "blue"

mycols[res_df$padj > 0.01 | is.na(res_df$padj)] <- "gray"

ggplot(res_df) +
  aes(x = log2FoldChange,
      y = -log(padj)) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log(0.01)) +
  ggtitle("Volcano Plot of Differential Expression")
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



The improved volcano plot highlights genes with large fold changes and strong statistical significance. The blue points are the genes with large expression changes and low adjusted p-values, so these are like the most important genes to focus on.

Adding Gene Annotation

Since the results currently only have Ensembl gene IDs, we will add gene symbols, Entrez IDs, and gene names.

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

Check the available annotation columns.

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

Q5. Use mapIds() to add SYMBOL, ENTREZID, and GENENAME annotation

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "SYMBOL",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = row.names(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID",
                     multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
res$name <- mapIds(org.Hs.eg.db,
                   keys = row.names(res),
                   keytype = "ENSEMBL",
                   column = "GENENAME",
                   multiVals = "first")
```

```
'select()' returned 1:many mapping between keys and columns
```

```
head(res, 10)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns
                   baseMean log2FoldChange     lfcSE       stat      pvalue
                  <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.913579      0.1792571 0.3248215   0.551863 5.81042e-01
ENSG00000187634  183.229650      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.188076     -0.6927205 0.0548465 -12.630156 1.43993e-36
ENSG00000187961  209.637938      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.255123      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.979750      0.5428105 0.5215598   1.040744 2.97994e-01
```

```
ENSG00000188290  108.922128      2.0570638 0.1969053  10.446970 1.51281e-25
ENSG00000187608  350.716868      0.2573837 0.1027266   2.505522 1.22271e-02
ENSG00000188157 9128.439422      0.3899088 0.0467164   8.346302 7.04333e-17
ENSG00000237330    0.158192      0.7859552 4.0804729   0.192614 8.47261e-01
                      padj      symbol      entrez                   name
                 <numeric> <character> <character>            <character>
ENSG00000279457 6.86555e-01          NA          NA                     NA
ENSG00000187634 5.15718e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76553e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13413e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19031e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03379e-01       PERM1       84808 PPARGC1 and ESRR ind..
ENSG00000188290 1.30538e-24        HES4       57801 hes family bHLH tran..
ENSG00000187608 2.37452e-02       ISG15        9636 ISG15 ubiquitin like..
ENSG00000188157 4.21970e-16        AGRN      375790                  agrin
ENSG00000237330          NA      RNF223      401934 ring finger protein ..
```

The annotation step adds more useful gene information to the DESeq2 results. This is important because pathway analysis uses Entrez gene IDs, and gene symbols/names make the results easier to understand.

Q6. Reorder results by p-value and save them to a CSV file

```r
res <- res[order(res$pvalue), ]

write.csv(res, file = "deseq_results.csv")
```

The results are now ordered by p-value and saved as `deseq_results.csv` in the project folder.

Section 2. Pathway Analysis

Now we will use the `gage` package for KEGG pathway analysis and `pathview` to draw pathway diagrams. This helps us understand which biological pathways are changing, instead of only looking at individual genes.

```r
library(pathview)
library(gage)
library(gageData)
```

Load the KEGG pathway datasets.

```r
data(kegg.sets.hs)
data(sigmet.idx.hs)

kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]

head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
```

```
[161] "9583"   "9615"
```

Create a named vector of fold changes. The values are the DESeq2 log2 fold changes, and the names are the Entrez gene IDs.

```
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez

head(foldchanges)
```

```
     1266     54855      1465      2034      2150      6659 
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288 
```

Run the GAGE pathway analysis.

```
keggres <- gage(foldchanges, gsets = kegg.sets.hs)

attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

Look at the top down-regulated pathways.

```
head(keggres$less)
```

```
                                        p.geomean stat.mean        p.val
hsa04110 Cell cycle                   8.995727e-06 -4.378644 8.995727e-06
hsa03030 DNA replication              9.424076e-05 -3.951803 9.424076e-05
hsa03013 RNA transport                1.375901e-03 -3.028500 1.375901e-03
hsa03440 Homologous recombination     3.066756e-03 -2.852899 3.066756e-03
hsa04114 Oocyte meiosis               3.784520e-03 -2.698128 3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis 8.961413e-03 -2.405398 8.961413e-03
                                            q.val set.size         exp1
hsa04110 Cell cycle                   0.001448312      121 8.995727e-06
hsa03030 DNA replication              0.007586381       36 9.424076e-05
hsa03013 RNA transport                0.073840037      144 1.375901e-03
hsa03440 Homologous recombination     0.121861535       28 3.066756e-03
hsa04114 Oocyte meiosis               0.121861535      102 3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis 0.212222694       53 8.961413e-03
```

Look at the top up-regulated pathways.

```r
head(keggres$greater)
```

```
                                      p.geomean stat.mean       p.val
hsa04640 Hematopoietic cell lineage   0.002822776  2.833362 0.002822776
hsa04630 Jak-STAT signaling pathway   0.005202070  2.585673 0.005202070
hsa00140 Steroid hormone biosynthesis 0.007255099  2.526744 0.007255099
hsa04142 Lysosome                     0.010107392  2.338364 0.010107392
hsa04330 Notch signaling pathway      0.018747253  2.111725 0.018747253
hsa04916 Melanogenesis                0.019399766  2.081927 0.019399766
                                          q.val set.size        exp1
hsa04640 Hematopoietic cell lineage   0.3893570       55 0.002822776
hsa04630 Jak-STAT signaling pathway   0.3893570      109 0.005202070
hsa00140 Steroid hormone biosynthesis 0.3893570       31 0.007255099
hsa04142 Lysosome                     0.4068225      118 0.010107392
hsa04330 Notch signaling pathway      0.4391731       46 0.018747253
hsa04916 Melanogenesis                0.4391731       90 0.019399766
```

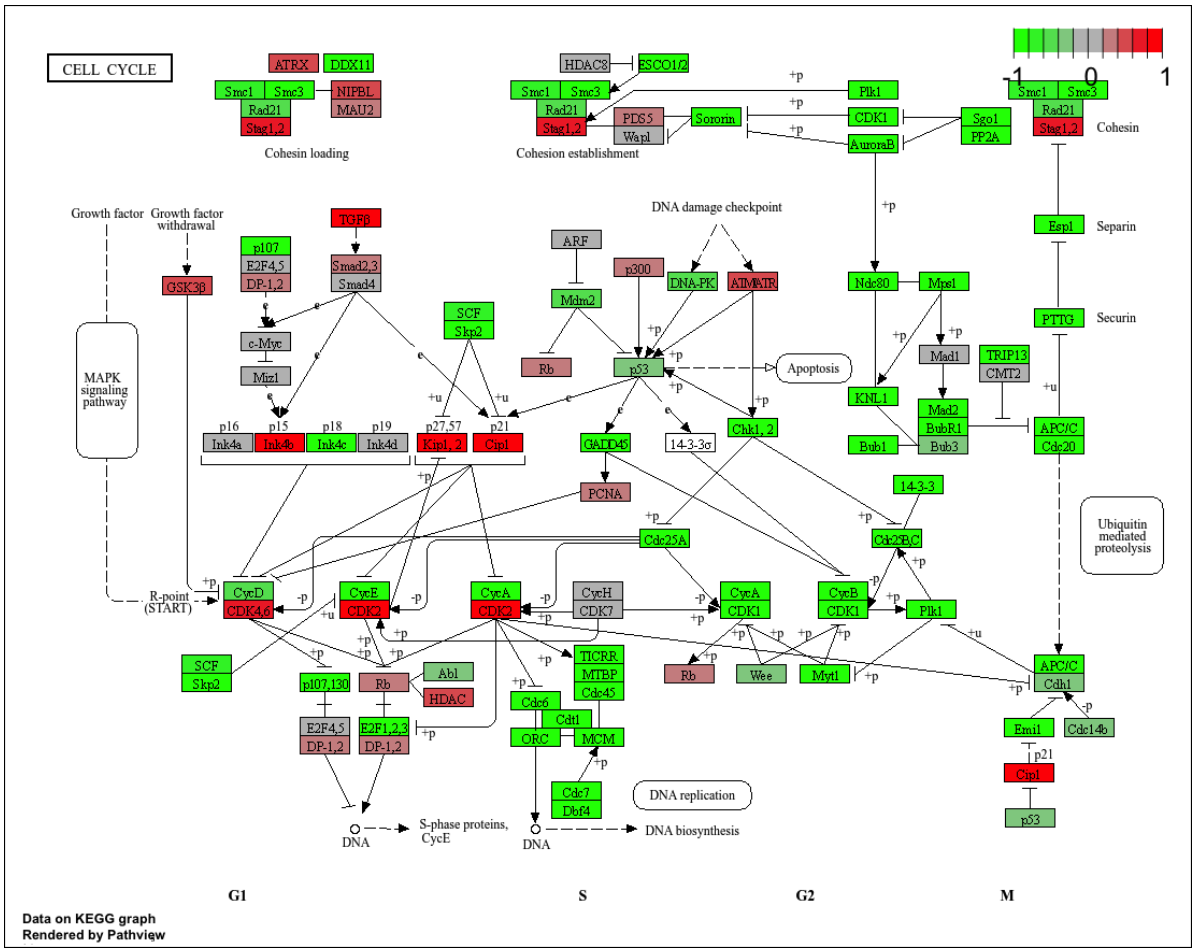
The top down-regulated pathways are expected to include cell cycle-related pathways. This makes sense because HOXA1 is connected to cell cycle progression.

Plot the Cell Cycle pathway.

```r
pathview(gene.data = foldchanges, pathway.id = "hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaifyenepoya/Desktop/BIMM143/LAB_14

Info: Writing image file hsa04110.pathview.png
```

Add the Cell Cycle pathway image to the report.



Q7. Can you do the same procedure to plot the pathview figures for the top 5 down-regulated pathways?

Get the top 5 down-regulated pathways.

```
down.pathways <- rownames(keggres$less)[1:5]
down.pathways
```

```
[1] "hsa04110 Cell cycle"                "hsa03030 DNA replication"
[3] "hsa03013 RNA transport"             "hsa03440 Homologous recombination"
[5] "hsa04114 Oocyte meiosis"
```

Extract the 8-character KEGG pathway IDs.

```r
down.ids <- substr(down.pathways, start = 1, stop = 8)
down.ids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

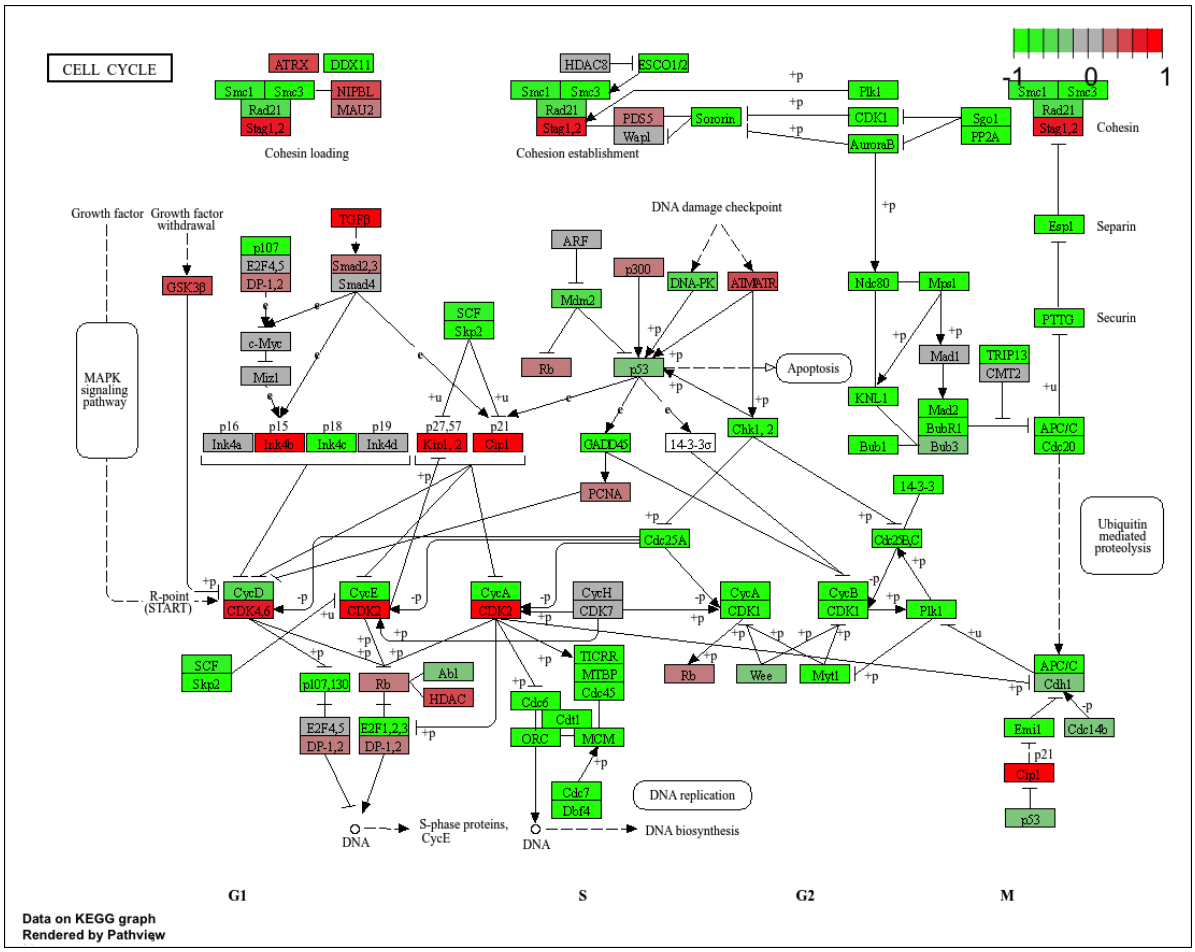
Use `pathview()` to draw the top 5 down-regulated pathway figures.

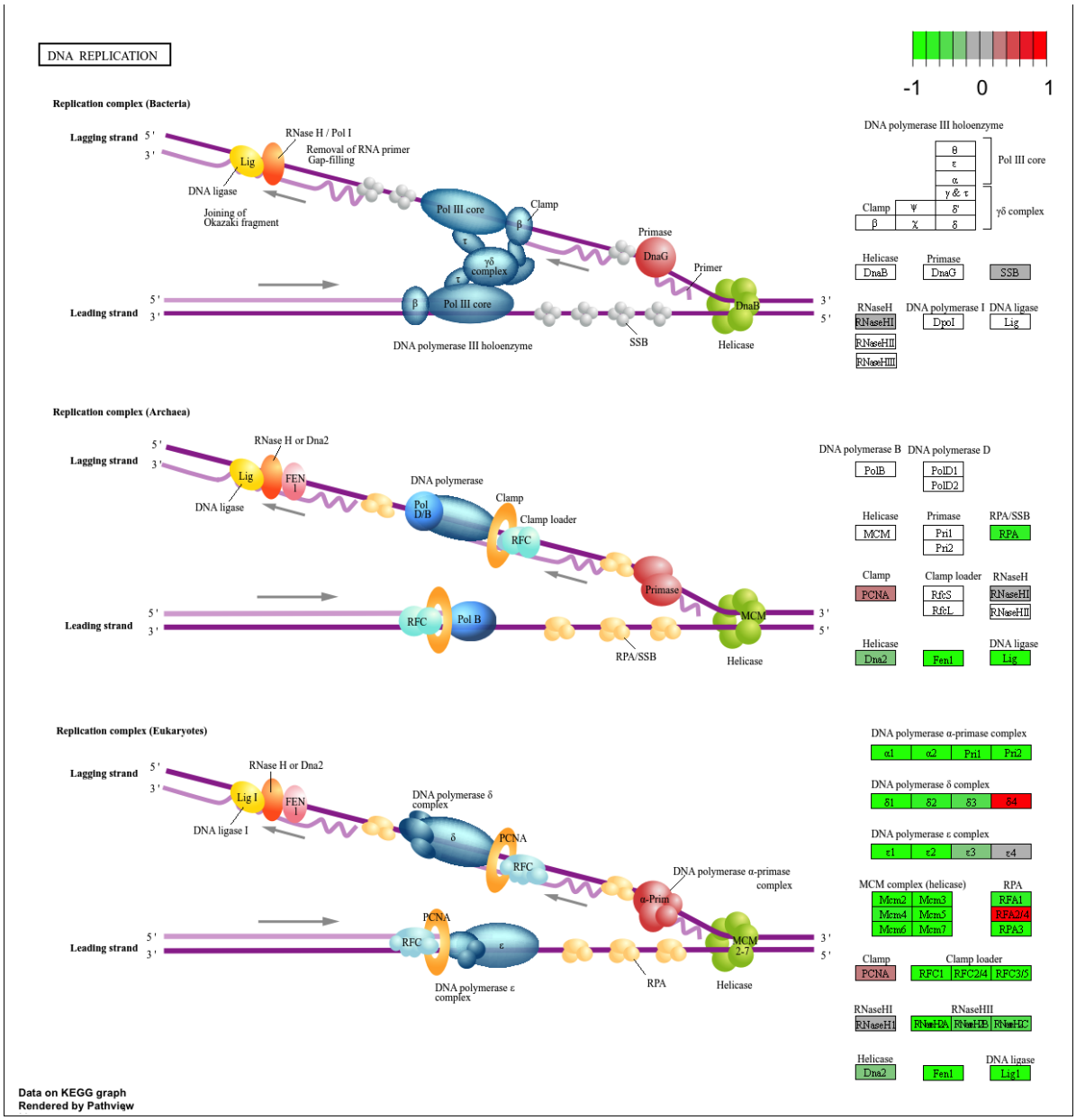
```r
pathview(gene.data = foldchanges, pathway.id = down.ids, species = "hsa")
```

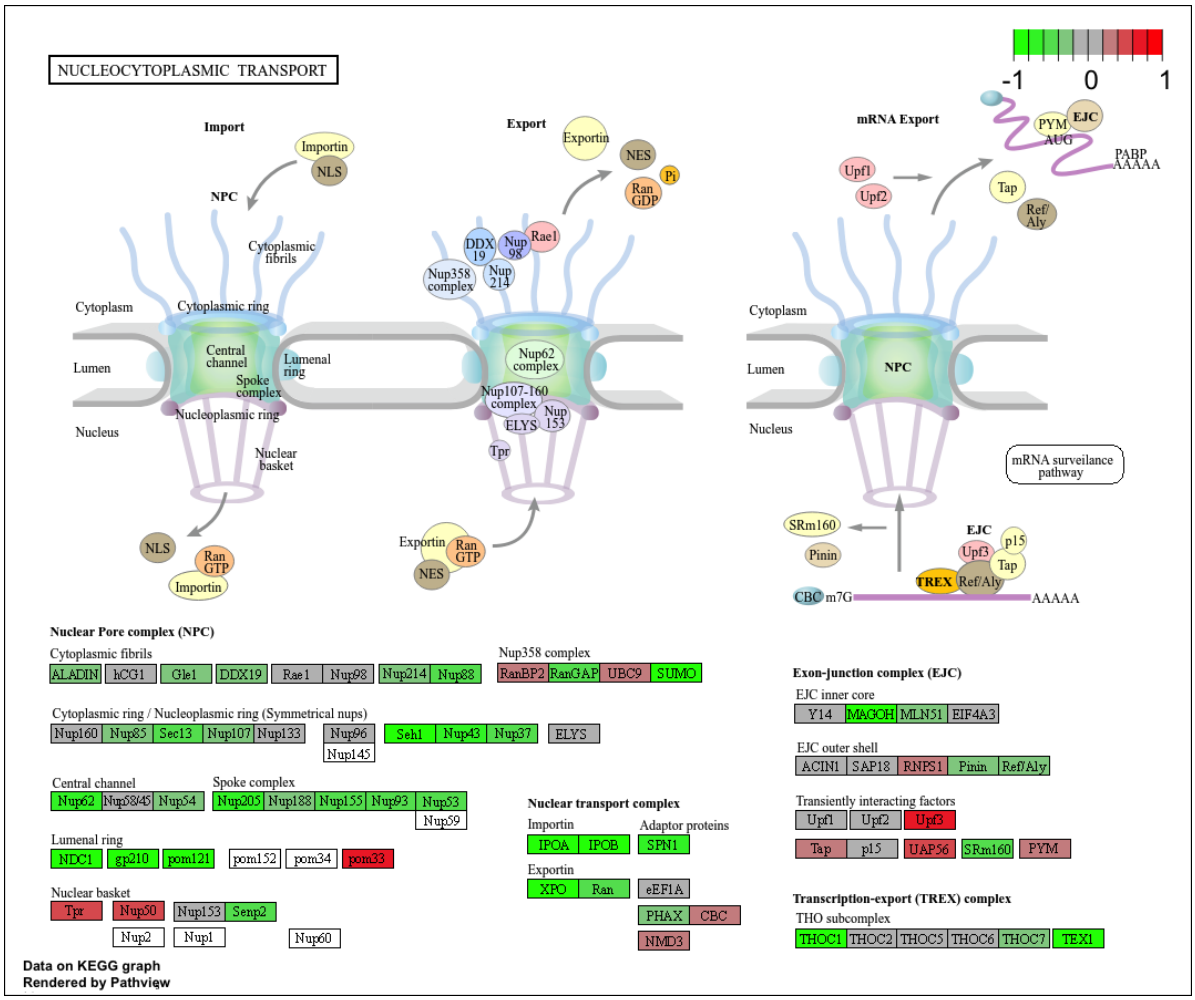
```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaifyenepoya/Desktop/BIMM143/LAB_14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaifyenepoya/Desktop/BIMM143/LAB_14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaifyenepoya/Desktop/BIMM143/LAB_14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaifyenepoya/Desktop/BIMM143/LAB_14

Info: Writing image file hsa03440.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaifyenepoya/Desktop/BIMM143/LAB_14
```

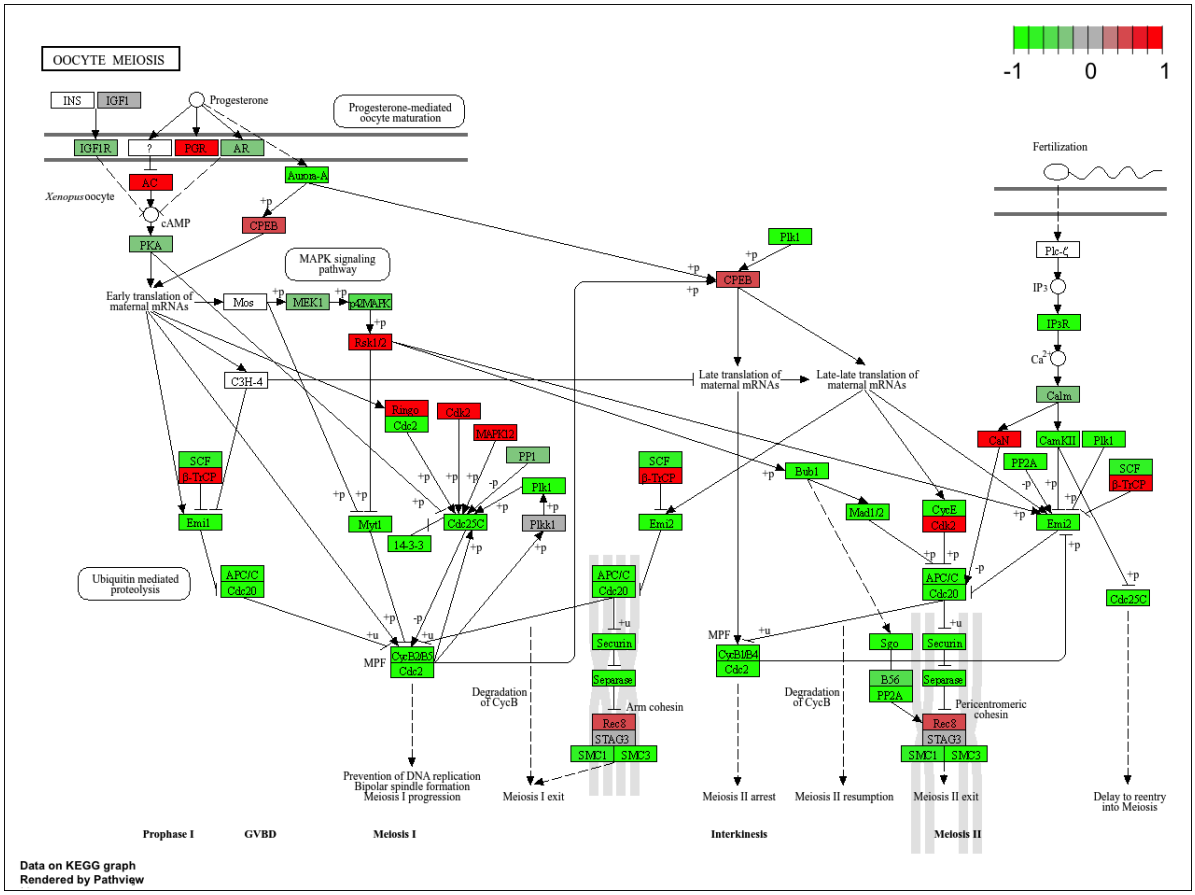
```
Info: Writing image file hsa04114.pathview.png
```

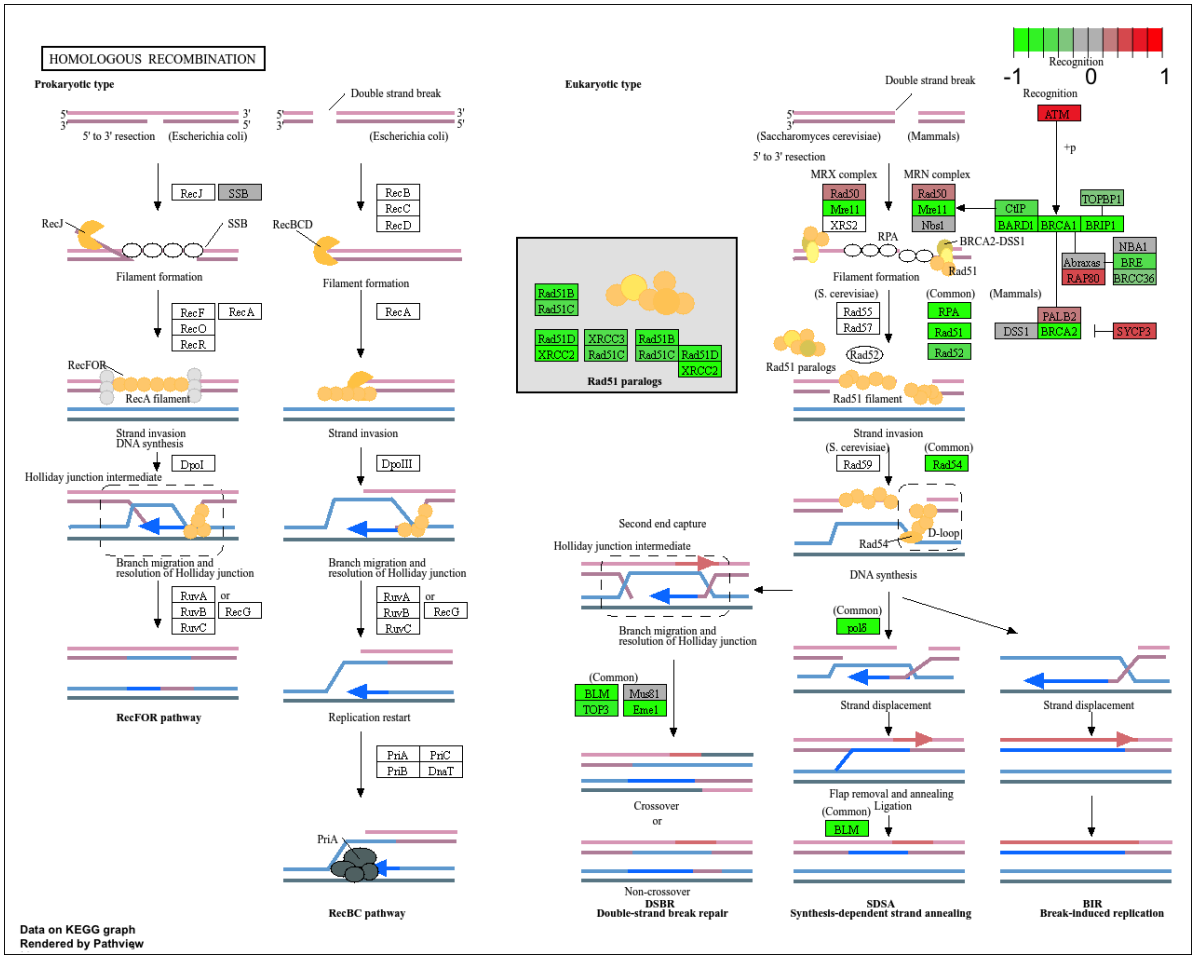
Add the pathway images to the report.




DNA REPLICATION
-1
0
1
Replication complex (Bacteria)
Lagging strand
5'
3'
RNase H / Pol I
Removal of RNA primer
Gap-filling
Lig
DNA ligase
Joining of
Okazaki fragment
Pol III core
Clamp
β
τ
γδ
complex
Primase
DnaG
Primer
Leading strand
DNA polymerase III holoenzyme
SSB
DnaB
Helicase
DNA polymerase III holoenzyme
θ
ε
α
γ & τ
Pol III core
Clamp
ψ
δ'
γδ complex
β
χ
δ
Helicase
DnaB
Primase
DnaG
SSB
RNaseH
RNaseHI
RNaseHII
RNaseHIII
DNA polymerase I
DpoI
DNA ligase
Lig
Replication complex (Archaea)
RNase H or Dna2
FEN
1
DNA polymerase
Pol
D/B
Clamp loader
RFC
Pol B
MCM
RPA/SSB
DNA polymerase B
PolB
DNA polymerase D
PolD1
PolD2
Helicase
MCM
Primase
Pri1
Pri2
RPA/SSB
RPA
Clamp
PCNA
Clamp loader
RfcS
RfcL
RNaseH
RNaseHI
RNaseHIII
Helicase
Dna2
Fen1
DNA ligase
Lig
Replication complex (Eukaryotes)
RNase H or Dna2
Lig I
FEN
DNA ligase I
DNA polymerase δ
complex
δ
PCNA
RFC
DNA polymerase α-primase
complex
α-Prim
ε
DNA polymerase ε
complex
MCM
2-7
RPA
Helicase
DNA polymerase α-primase complex
α1
α2
Pri1
Pri2
DNA polymerase δ complex
δ1
δ2
δ3
δ4
DNA polymerase ε complex
ε1
ε2
ε3
ε4
MCM complex (helicase)
Mcm2
Mcm3
Mcm4
Mcm5
Mcm6
Mcm7
RPA
RFA1
RFA2/4
RFA3
Clamp
PCNA
Clamp loader
RFC1
RFC2/4
RFC3/5
RNaseHI
RNaseHI1
RNaseHIII
RNaseHIA
RNaseHIB
RNaseHIC
Helicase
Dna2
Fen1
DNA ligase
Lig1
Data on KEGG graph
Rendered by Pathview


NUCLEOCYTOPLASMIC TRANSPORT
-1
0
1
Import
Importin
NLS
NPC
Cytoplasmic fibrils
Cytoplasm
Cytoplasmic ring
Central channel
Lumenal ring
Spoke complex
Lumen
Nucleoplasmic ring
Nucleus
Nuclear basket
NLS
Ran GTP
Importin
Export
Exportin
NES
Pi
Ran GDP
Rae1
DDX 19
Nup 98
Nup358 complex
Nup 214
Nup62 complex
Nup107-160 complex
ELYS
Nup 153
Tpr
Exportin
Ran GTP
NES
mRNA Export
EJC
PYM
AUG
PABP
AAAAA
Upf1
Upf2
Tap
Ref/ Aly
Cytoplasm
Lumen
NPC
Nucleus
mRNA surveilance pathway
SRm160
Pinin
EJC
p15
Upf3
Tap
TREX
Ref/Aly
CBC
m7G
AAAAA
Nuclear Pore complex (NPC)
Cytoplasmic fibrils
ALADIN
hCG1
Gle1
DDX19
Rae1
Nup98
Nup214
Nup88
Nup358 complex
RanBP2
RanGAP
UBC9
SUMO
Cytoplasmic ring / Nucleoplasmic ring (Symmetrical nups)
Nup160
Nup85
Sec13
Nup107
Nup133
Nup96
Nup145
Seh1
Nup43
Nup37
ELYS
Central channel
Nup62
Nup58/45
Nup54
Spoke complex
Nup205
Nup188
Nup155
Nup93
Nup53
Nup59
Lumenal ring
NDC1
gp210
pom121
pom152
pom34
pom33
Nuclear basket
Tpr
Nup50
Nup153
Senp2
Nup2
Nup1
Nup60
Nuclear transport complex
Importin
IPOA
IPOB
Adaptor proteins
SPN1
Exportin
XPO
Ran
eEF1A
PHAX
CBC
NMD3
Exon-junction complex (EJC)
EJC inner core
Y14
MAGOH
MLN51
EIF4A3
EJC outer shell
ACIN1
SAP18
RNPS1
Pinin
Ref/Aly
Transiently interacting factors
Upf1
Upf2
Upf3
Tap
p15
UAP56
SRm160
PYM
Transcription-export (TREX) complex
THO subcomplex
THOC1
THOC2
THOC5
THOC6
THOC7
TEX1
Data on KEGG graph
Rendered by Pathview


OOCYTE MEIOSIS
-1
0
1
INS
IGF1
Progesterone
Progesterone-mediated oocyte maturation
IGF1R
?
PGR
AR
AC
Xenopus oocyte
cAMP
PKA
Aurora-A
CPEB
MAPK signaling pathway
Early translation of maternal mRNAs
Mos
MEK1
c-MAPK
Rsk1/2
C3H-4
Ringo
Cdc2
Cdk2
MAPK1/2
PP1
Plk1
Plkk1
SCF
β-TrCP
Emi1
Myt1
14-3-3
Cdc25C
Ubiquitin mediated proteolysis
APC/C
Cdc20
MPF
CycB/B2
Cdc2
Degradation of CycB
Prevention of DNA replication
Bipolar spindle formation
Meiosis I progression
Securin
Separase
Arm cohesin
Rec8
STAG3
SMC1
SMC3
Meiosis I exit
Plk1
CPEB
Late translation of maternal mRNAs
Late-late translation of maternal mRNAs
SCF
β-TrCP
Emi2
Bub1
Mad1/2
CycE
Cdk2
Meiosis II arrest
CycB/B4
Cdc2
Sgo
B56
PP2A
Meiosis II resumption
Fertilization
Plc-ζ
IP3
IP3R
Ca2+
Calm
CaN
CamKII
Plk1
PP2A
SCF
β-TrCP
Emi2
APC/C
Cdc20
Cdc25C
Securin
Separase
Pericentromeric cohesin
Rec8
STAG3
SMC1
SMC3
Meiosis II exit
Delay to reentry into Meiosis
Prophase I
GVBD
Meiosis I
Interkinesis
Meiosis II
Data on KEGG graph
Rendered by Pathview



The same procedure was used to plot the top 5 down-regulated KEGG pathways. These pathways include Cell cycle, DNA replication, RNA transport, Oocyte meiosis, and Homologous recombination. This supports the idea that HOXA1 knockdown strongly affects cell cycle and DNA replication processes.

Section 3. Gene Ontology Analysis

We can also use GAGE for Gene Ontology analysis. Here we focus on Biological Process terms.

```
data(go.sets.hs)
data(go.subs.hs)

gobpsets <- go.sets.hs[go.subs.hs$BP]
```

Run GAGE analysis on Biological Process GO terms.

```r
gobpres <- gage(foldchanges, gsets = gobpsets)

lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.519724e-05  3.824205 8.519724e-05
GO:0002009 morphogenesis of an epithelium 1.396681e-04  3.653886 1.396681e-04
GO:0048729 tissue morphogenesis           1.432451e-04  3.643242 1.432451e-04
GO:0007610 behavior                       1.925222e-04  3.565432 1.925222e-04
GO:0060562 epithelial tube morphogenesis  5.932837e-04  3.261376 5.932837e-04
GO:0035295 tube development               5.953254e-04  3.253665 5.953254e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1951953      113 8.519724e-05
GO:0002009 morphogenesis of an epithelium 0.1951953      339 1.396681e-04
GO:0048729 tissue morphogenesis           0.1951953      424 1.432451e-04
GO:0007610 behavior                       0.1967577      426 1.925222e-04
GO:0060562 epithelial tube morphogenesis  0.3565320      257 5.932837e-04
GO:0035295 tube development               0.3565320      391 5.953254e-04

$less
                                             p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.536227e-15 -8.063910 1.536227e-15
GO:0000280 nuclear division              4.286961e-15 -7.939217 4.286961e-15
GO:0007067 mitosis                       4.286961e-15 -7.939217 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.169934e-14 -7.797496 1.169934e-14
GO:0007059 chromosome segregation        2.028624e-11 -6.878340 2.028624e-11
GO:0000236 mitotic prometaphase          1.729553e-10 -6.695966 1.729553e-10
                                                q.val set.size         exp1
GO:0048285 organelle fission             5.841698e-12      376 1.536227e-15
GO:0000280 nuclear division              5.841698e-12      352 4.286961e-15
GO:0007067 mitosis                       5.841698e-12      352 4.286961e-15
GO:0000087 M phase of mitotic cell cycle 1.195672e-11      362 1.169934e-14
GO:0007059 chromosome segregation        1.658603e-08      142 2.028624e-11
GO:0000236 mitotic prometaphase          1.178402e-07       84 1.729553e-10

$stats
                                           stat.mean     exp1
GO:0007156 homophilic cell adhesion         3.824205 3.824205
GO:0002009 morphogenesis of an epithelium   3.653886 3.653886
```

```
GO:0048729 tissue morphogenesis            3.643242 3.643242
GO:0007610 behavior                        3.565432 3.565432
GO:0060562 epithelial tube morphogenesis   3.261376 3.261376
GO:0035295 tube development                3.253665 3.253665
```

The GO results show enriched biological processes. The down-regulated GO terms are related to M phase, organelle fission, nuclear division, mitosis, and chromosome segregation. This matches the KEGG results because both show that cell cycle and cell division processes are strongly affected by HOXA1 knockdown.

Section 4. Reactome Analysis

Now we will create a plain text file of significant genes at the adjusted p-value 0.05 level. This file can be uploaded to Reactome for online pathway analysis.

```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]

print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```r
write.table(sig_genes,
            file = "significant_genes.txt",
            row.names = FALSE,
            col.names = FALSE,
            quote = FALSE)
```

The file `significant_genes.txt` is now saved in the project folder. This file can be uploaded to Reactome.

Q8. What pathway has the most significant Entities p-value? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The most significant Reactome pathway is related to cell cycle regulation, especially mitotic cell cycle and DNA replication. This matches the KEGG results because KEGG also showed Cell cycle and DNA replication as top affected pathways. Differences between Reactome and KEGG can happen because they use different pathway databases, pathway definitions, gene annotations, and statistical methods.

Final Interpretation

Overall, the DESeq2 results show that HOXA1 knockdown caused many gene expression changes. The pathway results strongly point toward changes in cell cycle, DNA replication, mitosis, and chromosome segregation. This supports the idea that HOXA1 is important for cell cycle progression in lung fibroblasts.

Session Information

```r
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sonoma 14.4.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats4    stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] gageData_2.48.0             gage_2.60.0
 [3] pathview_1.50.0             org.Hs.eg.db_3.22.0
 [5] AnnotationDbi_1.72.0        ggplot2_4.0.2
 [7] DESeq2_1.50.2               SummarizedExperiment_1.40.0
 [9] Biobase_2.70.0              MatrixGenerics_1.22.0
[11] matrixStats_1.5.0           GenomicRanges_1.62.1
[13] Seqinfo_1.0.0               IRanges_2.44.0
[15] S4Vectors_0.48.1            BiocGenerics_0.56.0
[17] generics_0.1.4
```

```
loaded via a namespace (and not attached):
 [1] KEGGREST_1.50.0     gtable_0.3.6        xfun_0.57
 [4] lattice_0.22-9      bitops_1.0-9        vctrs_0.7.2
 [7] tools_4.5.3         parallel_4.5.3      tibble_3.3.1
[10] RSQLite_3.52.0      blob_1.3.0          pkgconfig_2.0.3
[13] Matrix_1.7-4        RColorBrewer_1.1-3  S7_0.2.1
[16] graph_1.88.1        lifecycle_1.0.5     compiler_4.5.3
[19] farver_2.1.2        Biostrings_2.78.0   codetools_0.2-20
[22] htmltools_0.5.9     RCurl_1.98-1.18     yaml_2.3.12
[25] GO.db_3.22.0        pillar_1.11.1       crayon_1.5.3
[28] BiocParallel_1.44.0 DelayedArray_0.36.1 cachem_1.1.0
[31] abind_1.4-8         tidyselect_1.2.1    locfit_1.5-9.12
[34] digest_0.6.39       dplyr_1.2.1         labeling_0.4.3
[37] fastmap_1.2.0       grid_4.5.3          cli_3.6.5
[40] SparseArray_1.10.10 magrittr_2.0.5      S4Arrays_1.10.1
[43] XML_3.99-0.23       withr_3.0.2         scales_1.4.0
[46] bit64_4.8.0         rmarkdown_2.31      XVector_0.50.0
[49] httr_1.4.8          bit_4.6.0           png_0.1-9
[52] memoise_2.0.1       evaluate_1.0.5      knitr_1.51
[55] rlang_1.2.0         Rcpp_1.1.1          glue_1.8.0
[58] DBI_1.3.0           Rgraphviz_2.54.0    KEGGgraph_1.70.0
[61] jsonlite_2.0.0      R6_2.6.1
```